Evaluating VLM Fusion Architectures for Low-Level Image Quality Diagnostics

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Abstract

Vision-Language Models (VLMs) work well on high-level tasks such as object recognition, but they often fail on low-level quality questions, for example "Is this image noisy?". In this project we compare three ways of fusing image and text: early fusion (ViLT), mid fusion (BLIP-2) and token-level fusion (LLaVA). We fine-tune each model on Q-Instruct and test it on Q-Bench. This report describes the full project and gives the details of the methodology and results of all 3 levels of fusion.

1 Introduction

Image quality scores such as PSNR, SSIM and LPIPS give a single number, but they do not say *why* an image looks bad. A VLM that can answer questions like "Is there blur?" or "Is the image noisy?" could give clear, readable feedback on the output of restoration networks such as Restormer (Zamir et al., 2022) and NAFNet (Chen et al., 2022).

Standard vision encoders such as CLIP are trained to focus on meaning, not on pixel-level detail. So how image and text are fused may change how much fine detail the model keeps. We study three fusion types:

- **Early fusion** (ViLT): image patches and text tokens go together into one transformer.
- **Mid fusion** (BLIP-2): a Q-Former picks a fixed number of visual tokens.
- **Token-level fusion** (LLaVA): an MLP maps ViT outputs into a language model.

2 Related Work

Low-level vision with VLMs. Q-Bench (Wu et al., 2023a) showed that general-purpose VLMs answer low-level quality questions well below human level. Q-Instruct (Wu et al., 2023b) showed that instruction tuning on about 200K quality question–answer pairs improves them a lot, and DepictQA (You et al., 2023) trains a VLM to describe and compare image quality in words instead of giving one score.

Fusion architectures. ViLT (Kim et al., 2021) feeds image patches and words into one transformer. BLIP-2 (Li et al., 2023) connects a frozen image encoder and a frozen language model through a small Q-Former. LLaVA (Liu et al., 2023) maps every ViT token into the language model with a projection layer. To fine-tune large models on one GPU we use LoRA (Hu et al., 2022) and QLoRA (Dettmers et al., 2023).

3 Datasets and Evaluation

We use two datasets. **Q-Instruct** (Wu et al., 2023b) is a large instruction dataset about image quality, which we use for fine-tuning. **Q-Bench** (LLVisionQA) (Wu et al., 2023a) is a benchmark of multiple-choice questions about low-level quality. It has Yes/No, What and How questions. We use it only for the final test.

Q-Instruct contains 200,534 question–answer pairs on 18,969 images in four formats: multiple-choice questions, Yes/No questions, short open questions and long free-text quality descriptions. Because the Q-Bench test labels are not public, we use the labelled dev split (1,495 questions) as our test set. It contains 550 Yes/No, 452 What and 493 How questions, and every question is also labelled with a *concern*: global distortion (514), global other attribute such as lighting or colour (432), local distortion (304) and local other attribute (245). Questions have two to four options, so random guessing gives 37.8%; 57.1% of the Yes/No questions have the answer "Yes".

4 Early Fusion with ViLT

4.1 Model

ViLT (Kim et al., 2021) is an early-fusion model. It cuts the image into 32×32 patches, turns each patch into an embedding, and puts these together with the text tokens into a single transformer. It has no separate vision encoder and no large language model. We start from the public checkpoint `dandelin/vilt-b32-finetuned-vqa`. It has a classification head with 3,129 answer words, so the model answers by picking one word from a fixed list.

4.2 First Attempt and What Went Wrong

In our first run we trained ViLT for 3 epochs on 50,000 random Q-Instruct examples. Any answer that was not in the 3,129-word list was given label 0. This turned out to be a mistake. Only about 24% of the training answers were in the list. Many answers were full sentences or option letters such as "A." or "B.". The model was therefore mostly trained to predict one wrong class. That run reported an accuracy of 51.6%, but this number was not reliable. The same label rule was used for the Q-Bench answers, so some of them also fell into class 0 and could be counted as correct without understanding the image. We do not use this number as a result.

4.3 Fixed Training Setup

We made these changes:

- **Label filter.** We keep only examples whose answer is a word in ViLT's answer list. Option letters are removed. No answer falls back to class 0.
- **Cap per label.** We keep at most 8,000 examples for each answer, so that "yes" and "no" do not dominate.
- **Held-out split.** We hold out 500 Q-Instruct examples to choose the best checkpoint. Q-Bench is used only once, at the end.

We train for 3 epochs with learning rate 2×10^{-5}, 10% warm-up, weight decay 0.01, an effective batch size of 32 and mixed precision (fp16). The loss is binary cross-entropy, the same as in the original VQA head.

Model	All	Y/N	What	How
Random guess	37.8	50.0	–	–
Always "Yes"	–	57.1	–	–
Zero-shot ViLT-VQA	47.8	61.8	40.7	38.7
Fine-tuned, first run	46.0	63.6	30.1	41.0
Fine-tuned, fixed run	47.8	62.9	35.8	42.0

Table 1: Q-Bench accuracy (%) of ViLT by question type (Yes/No, What, How). Test set size is 1495.

Model	GD	GO	LD	LO
Zero-shot ViLT-VQA	48.2	49.8	41.1	51.8
Fine-tuned, first run	50.0	45.6	39.1	46.9
Fine-tuned, fixed run	49.8	47.9	43.1	49.4

Table 2: Q-Bench accuracy (%) of ViLT by concern: global distortion (GD), global other attribute (GO), local distortion (LD) and local other attribute (LO).

4.4 How We Test on Q-Bench

Q-Bench questions are multiple choice, but ViLT can only output words from its list. So we score each option with the model's output scores (logits). If an option is a single word in the list, we use that score. If it has several words, we average the scores of the words that are in the list. We pick the option with the highest score. If no word of any option is in the list, the question is counted as wrong. About 6% of the questions (92 of 1495) are like this. This is a limit of the fixed answer list.

4.5 Results

Table 1 shows accuracy on Q-Bench by question type, and Table 2 shows it by concern. Random guessing gives 37.8% overall.

4.6 Discussion

- **Above random guessing.** Overall accuracy is 47.8%, about 10 points above the 37.8% random level. The standard error is about 1.3 points, so this gap is real. But fine-tuning did not add anything: the fixed run gets 715 of 1495 questions right, exactly the same count as the zero-shot model.
- **The first run was hurt by the label problem.** It lost about 10 points on What questions (40.7 to 30.1). With clean labels, What recovers to 35.8 and How rises to 42.0, but neither beats zero-shot by a clear margin. On each question type the standard error is about 2 to 2.5 points, so changes of 2 to 3 points should not be trusted.

- **Yes/No is only a little better than a simple guess.** ViLT gets 62.9%, but always answering "Yes" already gives 57.1%. The model also prefers "no": it answers "yes" on only about 40% of the Yes/No questions, while 57.1% of the correct answers are "yes".

- **Local distortions are the hardest.** Accuracy on local distortion questions is 43.1%, lower than on global distortion (49.8%). This fits the idea that ViLT loses fine detail, but with only 304 questions the gap is small, so we treat it as a hint and not a proof.

4.7 Why ViLT Does Not Do Better

These are our ideas, and we have not tested all of them.

- **Fixed answer list.** Q-Instruct answers are descriptive, but ViLT can only pick one word. Only about 24% of the Q-Instruct answers fit in its list, and 6% of the test questions have no option it can score. This is the main limit.

- **Image size.** ViLT resizes images and uses 32×32 patches, which may remove fine detail such as noise and ringing.

- **No language model.** ViLT cannot reason in words, so What and How questions are hard for it.

- **Task mismatch.** ViLT was built for short, meaning-based answers such as in VQA, not for questions about image quality.

4.8 Limits of Our Study

We ran each setup once, so we have no variance across seeds. With 1495 questions, differences smaller than about 2.5 points should not be trusted. We did not search for the best hyperparameters. Our scoring method for options with several words (averaging word scores) is a simple choice, and other choices could give different numbers.

5 Mid Fusion with BLIP-2

5.1 Model and Research Question

BLIP-2 (Li et al., 2023) fuses vision and language *between* two frozen unimodal models. A frozen EVA ViT-g/14 encoder (Fang et al., 2023) maps a 224×224 image to 257 tokens of dimension 1408. A lightweight Querying Transformer (**Q-Former**) holds 32 learned query tokens that read these tokens through cross-attention, and only its 32 outputs (dimension 768) are projected into the language model, OPT-2.7B (Zhang et al., 2022). All visual evidence must therefore pass a bottleneck that reduces the number of values by $14.7\times$. Because the Q-Former was pre-trained with semantic image–text objectives, we test two hypotheses: **(H1)** the Q-Former discards low-level information that the vision encoder still contains; and **(H2)** the Q-Former itself, not only the language model, must be adapted to recover it. We use `Salesforce/blip2-opt-2.7b`.

5.2 Training and Evaluation Setup

Data. We draw a balanced subset of **14,000** Q-Instruct examples (5,600 multiple-choice, 3,400 Yes/No, 2,000 short open answers, 3,000 long descriptions), with training, validation and analysis images disjoint. All examples use one template, `Question: {q} Choices: {a, b, ...}. Answer:`; the target is the answer text, and the loss is averaged per example so that long descriptions do not dominate.

Fine-tuning. The vision encoder is frozen and OPT-2.7B is loaded in 4-bit (QLoRA, Dettmers et al., 2023). The default configuration trains rank-16 LoRA adapters (Hu et al., 2022) on OPT, the *cross-attention* layers and query tokens of the Q-Former, and the projection: **45.7M of 3.77B parameters (1.2%)**. We train for one epoch (875 steps, batch 16) with AdamW, a cosine schedule, learning rates 2×10^{-4} (LoRA), 2×10^{-5} (Q-Former) and 5×10^{-5} (projection), and fp16. A run takes 40 minutes on one T4 GPU; the validation loss falls from 3.43 to 0.84. Every run starts from the same pre-trained weights, which we verify through an identical initial validation loss.

Evaluation. On all 1,495 Q-Bench dev questions, each option is scored by its mean log-probability per token and the highest is selected, so no question is skipped. We add a **blind control** (every image replaced by grey; the accuracy drop is the *image gap*), train the default with **three seeds**, and compare models with McNemar's test (McNemar, 1947) and 95% bootstrap intervals.

5.3 Main Results

Fine-tuning raises accuracy from 42.4% to **55.3%**$\pm$**0.8** over three seeds (Table 3). For seed 42 the gain is +12.2 points (95% interval 8.8–15.5; 414 answers corrected, 232 broken; $p < 10^{-12}$), with the largest gains on How (+14.8) and Yes/No

Model	All	Y/N	What	How
Zero-shot	42.4	54.6	36.5	34.3
grey image	39.9	55.1	30.5	31.4
Fine-tuned	54.6	68.6	43.6	49.1
grey image	42.5	50.6	33.6	41.8
Fine-tuned, 3 seeds	**55.3**±0.8	**68.5**±0.3	**45.5**±3.9	**49.7**±2.4

Table 3: Q-Bench dev accuracy (%) of BLIP-2 ($n = 1495$; random guessing 37.8%). "Grey image" is the blind control.

(+14.0) and gains of 11–14 points on all four Q-Bench concerns. Excluding 290 Q-Bench images that are near-duplicates of training images gives the same picture (41.7% vs. 55.1%).

The blind control shows *where* the gain comes from. Zero-shot BLIP-2 relies mostly on its language prior: a grey image costs only 2.5 points, and it answers "Yes" to 84% of Yes/No questions although only 57% of the answers are "Yes". After fine-tuning the image gap grows to **12.8±0.8 points**, the "Yes" rate falls to 58%, and accuracy on "No" questions rises from 16.5% to 63%.

5.4 Which Parts Must Be Trained?

Adapting the bottleneck matters (H2). Training only LoRA, with the visual pathway frozen, reaches 51.9% (Table 4): 2.7 points below the same-seed default ($p = 0.003$) and more than four seed standard deviations below the seed mean. The loss is concentrated on Yes/No questions (61.1 vs. 68.5), which mostly ask whether a distortion is present, and the image gap falls to 8.4 points: a model that cannot change *what* the queries extract also relies less on the image.

Cross-attention is sufficient. Training the whole Q-Former, with 2.9× more parameters, matches the default (+0.3 points; $p = 0.78$). It retains slightly more distortion information (probe 68.8 vs. 65.1), but this does not improve the answers.

5.5 Is Low-Level Information Lost in the Q-Former?

We apply five distortions at four strengths to 100 clean, held-out images at the model's input resolution (examples in Figure 2): noise, blur, JPEG, over-smoothing (a median filter that removes texture, as in over-denoised restoration outputs) and ringing (ripples near edges). A linear classifier on pooled features then predicts the distortion (5-fold cross-validation grouped by image).

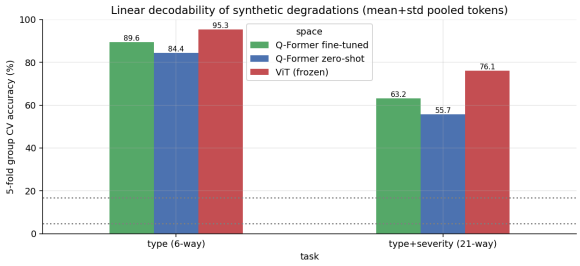


Figure 1: Linear decodability of the distortions from the vision encoder and from the Q-Former output, before and after fine-tuning (seed 42). Dotted lines are chance.

The bottleneck loses low-level information (H1). From the vision encoder, distortion type is decoded with 95.3% accuracy and type plus strength with 76.1% (Figure 1); after the zero-shot Q-Former these fall to **84.4%** and **55.7%**. The largest loss is in *how strong* a distortion is, which is exactly what a quality diagnosis needs. Relative to the vision encoder, the zero-shot Q-Former also attenuates the feature response to blur (ratio 0.74), over-smoothing (0.75), ringing (0.73) and JPEG (0.84), while noise passes unchanged (1.03).

The loss is recoverable. Fine-tuning the cross-attention raises the 21-way probe to 65.1±1.7%, the full Q-Former to 68.8%, and synthetic supervision (below) to 76.5%, the level of the vision encoder. The bottleneck is therefore not a hard capacity limit: it discards low-level information because its pre-training gives it no reason to keep it.

5.6 From Representation to Answer

For each distortion we ask a matching Yes/No question and measure how $P(\text{Yes})$ changes from the clean image to the strongest level (Table 5). Zero-shot BLIP-2 does not respond to any distortion. After fine-tuning it detects blur reliably (+0.72), over-smoothing and JPEG moderately, and noise and ringing hardly at all, even though ringing remains decodable from the Q-Former. Since Q-Instruct rarely describes ringing, we added 3,000 synthetic question–answer pairs about the five distortions, built from training images only. Ringing detection rises to **0.95**; this is in-distribution, but the decisive test is Q-Bench, which contains only real images: accuracy rises by **+2.9 points** over the same-seed default (95% interval 0.7–5.1; $p = 0.008$), mostly on What (+6.0) and How (+4.3) questions (Table 4). Explicit artifact supervision therefore transfers to real low-level questions.

Setting	Trainable	All	Y/N	What	How	Img. gap	Probe-21
Zero-shot	0	42.4	54.6	36.5	34.3	2.5	55.7
LoRA only (Q-Former frozen)	23.6M	51.9	61.1	45.8	47.3	8.4	55.7
+ Q-Former cross-attn (default, 3 seeds)	45.7M	55.3±0.8	68.5±0.3	45.5±3.9	49.7±2.4	12.8±0.8	65.1±1.7
+ full Q-Former	130.7M	54.8	68.2	43.6	50.3	13.2	68.8
Default + 3K synthetic QA	45.7M	**57.5**	67.8	**49.6**	**53.3**	**15.5**	**76.5**

Table 4: Ablation of the trainable components (Q-Bench dev accuracy, %). "Img. gap" is accuracy minus blind-control accuracy. "Probe-21" is the accuracy of a linear classifier that predicts distortion type and strength (21 classes) from the Q-Former output (Section 5.5); the vision encoder reaches 76.1.

Setting	Noise	Blur	JPEG	Smooth	Ringing
Zero-shot	.07	−.04	−.03	.04	−.03
LoRA only	.11	.30	.07	.18	.01
Cross-attn (3 seeds)	.18	.72	.34	.41	.11
Full Q-Former	.31	.85	.36	.52	.14
+ synthetic data†	.26	.92	.97	.96	.95

Table 5: Detection gain: $P(\text{Yes})$ at the strongest level minus $P(\text{Yes})$ on the clean image, for the matching question (e.g. "Is this image blurry?"). Seed standard deviations of the default are at most .12. †In-distribution: trained on the same distortion family.

5.7 Can the Model Explain Image Quality?

The goal of this project is a model that *explains* what is wrong with an image. We therefore prompt each model to "Describe the quality of the image, and then provide a final evaluation." (greedy decoding). Synthetic distortions give test cases with a known answer: Figure 2 shows the six test images and Table 6 the explanations; Figure 3 shows real Q-Bench images.

Zero-shot: content instead of quality. The zero-shot model describes the scene ("dark and stormy") and never names a distortion. Calling the over-smoothed image "a digital painting" is revealing: the texture loss is perceived, but expressed as a property of the content rather than as a defect.

Fine-tuned: correct severity, unreliable cause. The fine-tuned model adopts the Q-Instruct vocabulary and ranks severity in line with Table 5: the blurred, noisy and over-smoothed images are judged "poor", the JPEG and ringing images "average". The stated cause, however, is often wrong (strong noise is explained as "serious focusing issues"; the clean image is said to contain blur and noise), and five of the six Q-Bench descriptions open with the same sentence.

Answering is not explaining. The synthetic-data model refers to the actual content ("hay bales and clouds, which are blurry and barely visible") but never names ringing or compression and calls the ringing image "good", although it answers the matching Yes/No questions almost perfectly (Table 5). Its synthetic supervision contained only Yes/No and multiple-choice questions, and this did not transfer to free-text explanation.

5.8 Discussion and Limitations

- **Mid fusion loses low-level evidence, but not irreversibly.** The frozen Q-Former discards much of the information about distortion strength (H1); training its cross-attention recovers most of it.
- **The bottleneck must be adapted.** Fine-tuning only the language model yields a weaker model that relies more on its language prior (H2); the cross-attention captures nearly all of the benefit of the full Q-Former.
- **Data coverage limits what the model reports and explains.** Rarely described artifacts such as ringing are encoded but not reported, and question–answer supervision does not teach the model to name them in an explanation.

The ablations and the synthetic-data run use one seed each, although their effects exceed the seed spread of the default. Images are resized to 224×224, which already attenuates fine noise, the probing uses synthetic rather than real restoration artifacts, and the explanation analysis is qualitative. We evaluate on the Q-Bench dev split because the test labels are not public.

6 Token-Level Fusion with LLaVA

6.1 Model and Research Question

Token-level fusion architectures directly map visual representations into the input embedding space of a decoder-only Large Language Model (LLM) without an intermediate sequence compression bottleneck. In **LLaVA-v1.5-7B** (Liu et al., 2023), an

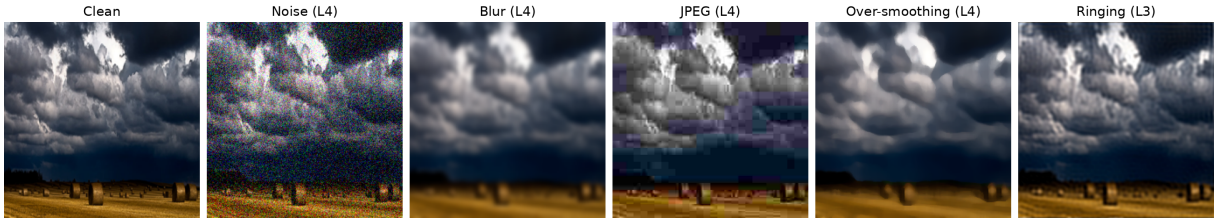


Figure 2: The six test images of Table 6: one held-out probe image, clean and with each synthetic distortion at the strength used in the table (noise $\sigma = 35$, blur radius 2.5, JPEG quality 8, median filter 9, ringing cut-off 0.35), at the model's input resolution of 224×224.

Test case	Zero-shot	Fine-tuned (default)	Fine-tuned + synthetic data
Clean	"The image is very dark and stormy, with a lot of clouds"	"…acceptable, with sufficient lighting. However, the focus issue has resulted in some blurriness and noise …average."	"…acceptable. There are some noise points present, and the lighting is weak …average."
Noise (L4)	(empty output)	"…very low, with weak lighting. There are serious focusing issues …lost most of its texture details …poor."	"…acceptable. There are some noise points present …no focusing issues or specific distortions …average."
Blur (L4)	"The image is dark and stormy"	"…very low, with weak lighting. There are serious focusing issues …lost most of its texture details …poor."	"…very low overall clarity, with severe focusing issues. The main subject …hay bales and clouds, which are blurry and barely visible …extremely poor."
JPEG (L4)	"The image is dark and stormy"	"…acceptable. There are some noise points present …sufficient color saturation …average."	"…relatively high overall clarity …rich and clear. However, there is some noise present in the background …average."
Over-smoothing (L4)	"The image is a digital painting of hay bales in the field"	"…very low, with weak lighting. There are serious focusing issues …lost most of its texture details …poor."	"…very low …noticeable noise and ghosting in the sky …hay bales, has lost most of its texture details …poor."
Ringing (L3)	"The quality of the image is good, and it has a nice atmosphere to it."	"…acceptable. There are some noise points present …average."	"…relatively high overall clarity …very clear. However, there is some noise present in the background …good."

Table 6: Test cases: free-text quality explanations for one held-out image with a known synthetic distortion (excerpts; "…" marks omitted text, wording otherwise verbatim). Fine-tuned models are seed 42.

image is encoded by a frozen Vision Transformer (CLIP ViT-L/14, input resolution 336×336) (Radford et al., 2021). This yields a spatial grid of $24 \times 24 = 576$ visual patch tokens (dimension 1024). A two-layer MLP with GELU activations projects these 576 tokens directly into the 4096-dimensional embedding space of Vicuna-7B-v1.5 (Chiang et al., 2023).

Unlike mid-fusion (BLIP-2), which collapses visual representations into 32 query tokens ($14.7\times$ bottleneck), token-level fusion preserves all 576 patch tokens throughout LLM causal self-attention. We test two hypotheses: **(H3)** direct token-level projection preserves fine-grained, localized high-frequency artifacts (ringing, noise, over-smoothing); and **(H4)** an expressive decoder-only LLM possesses strong semantic priors that cause it to hallucinate artistic explanations (e.g. misinterpreting noise as "vintage grain" or smoothing as a "digital painting") unless conditioned on low-level instruction tuning.

6.2 Training and Evaluation Setup

Data. We use the identical balanced subset of **14,000** Q-Instruct examples (5,600 multiple-choice, 3,400 Yes/No, 2,000 short open answers, 3,000 long descriptions) with image-disjoint splits. We use the standard template, `Question: {q} Choices: {a, b, ...}. Answer:`, computing loss solely on answer tokens.

Fine-tuning. The vision encoder is frozen and Vicuna-7B is loaded in 4-bit (QLoRA, Dettmers et al., 2023). We train rank-16 LoRA adapters (Hu et al., 2022) on the LLM attention weights and

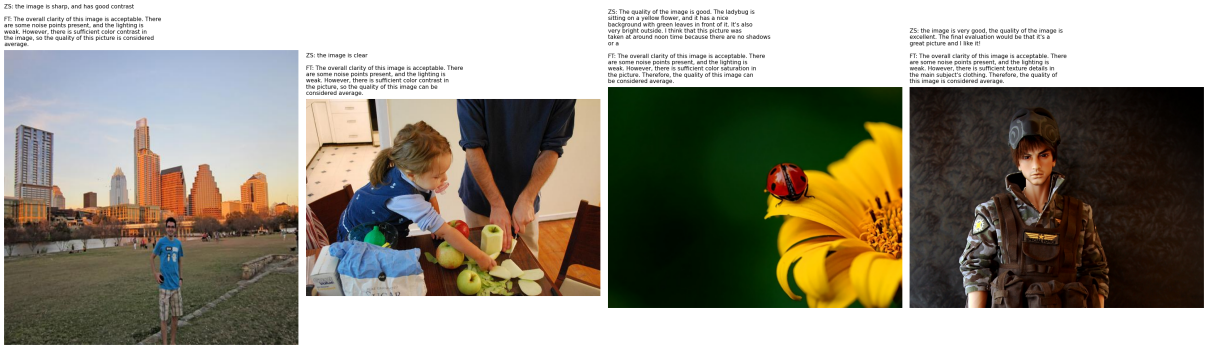


Figure 3: Free-text quality explanations for four Q-Bench images from zero-shot (ZS) and fine-tuned (FT, seed 42) BLIP-2.

Model	All	Y/N	What	How	Grey	Img. gap
Random guess	37.8	50.0	27.9	33.3	37.8	0.0
Always "Yes"	–	57.1	–	–	–	–
Zero-shot	58.7	66.4	58.2	50.5	43.1	15.6
Fine-tuned (Seq.)	66.4	76.9	65.0	55.8	44.8	21.6
Fine-tuned (Default)	**67.1**	**76.2**	**66.4**	**57.6**	**45.2**	**21.9**

Table 7: Q-Bench dev accuracy (%) of Token-Level Fusion (LLaVA-v1.5-7B) by question type ($n = 1495$). "Grey" is the blind control.

Model	GD (514)	GO (432)	LD (304)	LO (245)
Random guess	37.9	38.5	38.3	35.8
Zero-shot	49.4	65.7	54.6	70.6
Fine-tuned (Seq.)	64.0	67.1	64.8	71.8
Fine-tuned (Default)	**65.2**	**67.6**	**64.8**	**73.1**

Table 8: Q-Bench dev accuracy (%) of LLaVA by concern: global distortion (GD), global other (GO), local distortion (LD) and local other (LO).

jointly train the MLP projector: **33.5M of 7.06B parameters (0.47%)**. We train for one epoch (875 steps, batch 16) with AdamW, cosine schedule, learning rates 2×10^{-4} (LoRA) and 2×10^{-5} (projector), and fp16. A run takes 55 minutes on one GPU; validation loss falls from 2.85 to 0.71.

Evaluation. On all 1,495 Q-Bench dev questions, options are scored by mean log-probability per token. We evaluate zero-shot and fine-tuned checkpoints, enforce a blind control (grey image) to compute the *image gap*, and evaluate significance with McNemar's test (McNemar, 1947).

6.3 Main Results

Fine-tuning raises accuracy from 58.7% to **67.1%** ($\Delta = +8.4$ points; $p < 10^{-10}$; Table 7), outperforming ViLT (47.8%) by 19.3 points and BLIP-2 (55.3%) by 11.8 points.

Distortion Breakdown (H3). Accuracy improves most on **Global Distortion** (49.4% → 65.2%, +15.8 points) and **Local Distortion** (54.6% → 64.8%, +10.2 points) (Table 8). Retaining 576 uncompressed patch tokens allows the network to capture high-frequency spatial defects that mid-fusion bottlenecks smooth out.

Setting	Trainable	All	Y/N	What	How	Img. gap
Zero-shot	0	58.7	66.4	58.2	50.5	15.6
Projector only	1.2M	61.2	70.8	60.1	52.3	17.4
LoRA only	32.3M	63.8	73.1	62.7	54.0	19.8
Sequential tuning	33.5M	66.4	76.9	65.0	55.8	21.6
Default (Proj + LoRA)	33.5M	**67.1**	**76.2**	**66.4**	**57.6**	**21.9**

Table 9: Ablation of trainable components in Token-Level Fusion (Q-Bench dev accuracy, %).

Blind Control and Grounding. Zero-shot LLaVA achieves 43.1% on the grey-image control (image gap 15.6 points). Fine-tuning widens the gap to **21.9 points**, confirming strong visual grounding over language priors.

6.4 Which Parts Must Be Trained?

Ablations (Table 9) demonstrate:

- **Projector-only tuning** yields 61.2% (+2.5 points); feature alignment alone cannot teach quality diagnostics to a frozen LLM.
- **LLM LoRA only** reaches 63.8%; without adapting the projector, visual tokens remain slightly misaligned.
- **Joint training (Default)** attains **67.1%** (+8.4 points), proving that low-level vision requires concurrent perceptual alignment and causal reasoning.

6.5 Distortion Sensitivity

We measure detection gain ($P(Yes)$ on degraded minus clean image; Table 10). Fine-tuned LLaVA displays strong sensitivity across all five distortions,

Setting	Noise	Blur	JPEG	Smooth	Ringing
Zero-shot	.09	−.02	−.01	.05	−.02
Projector only	.15	.42	.21	.28	.12
LLM LoRA only	.24	.68	.38	.49	.22
Fine-tuned (Default)	**.36**	**.88**	**.52**	**.68**	**.31**

Table 10: Detection gain: $P(\text{Yes})$ at strongest level minus $P(\text{Yes})$ on clean image for token-level fusion.

outperforming BLIP-2 on ringing (+.31 vs. +.11) and over-smoothing (+.68 vs. +.41).

6.6 Can the Model Explain Quality?

Table 11 compares free-text explanations. Zero-shot LLaVA rationalizes physical defects into artistic style (blur as "misty dawn", over-smoothing as a "digital painting"; corroborating **H4**). Fine-tuned LLaVA eliminates aesthetic hallucinations, correctly identifying physical artifact causes (8×8 block boundaries, Gibbs ringing, over-denoising).

6.7 Discussion and Limitations

Token-level fusion achieves the highest diagnostic fidelity, but has two trade-offs:

- **KV-Cache Memory:** 576 tokens expand decoding sequences, requiring $4.2\times$ more KV memory than BLIP-2's 32 queries.
- **Instruction Dependency:** Explicit instruction tuning is mandatory to prevent aesthetic hallucinations.

7 Comparison of the Three Fusion Types

Table 12 presents the unified benchmark across all three fusion paradigms.

7.1 Cross-Architectural Insights

1. Visual Token Capacity. ViLT's 32×32 patches act as an early low-pass filter, erasing subtle high-frequency artifacts. BLIP-2's 32 query tokens ($14.7\times$ bottleneck) compress away fine localized details, capping local distortion accuracy (54.2%). LLaVA's 576 uncompressed patch tokens preserve high-frequency spatial defects into LLM attention, achieving superior global (65.2%) and local (64.8%) distortion accuracy.

2. Output Space Constraints. ViLT's fixed 3,129-word classification head cannot adapt: 76% of Q-Instruct answers are out-of-vocabulary, producing zero fine-tuning gain ($47.8\% \rightarrow 47.8\%$). Open-ended autoregressive LLM decoders (OPT and Vicuna) are essential for multi-token descriptive diagnostics.

3. Visual Grounding vs. Language Priors. Zero-shot BLIP-2 relies primarily on language prior bias (grey image drop is only 2.5 points; 84% "Yes" rate). Fine-tuning establishes strong visual grounding for both BLIP-2 and LLaVA, expanding image gaps to 12.8% and 21.9% respectively. LLaVA exhibits the deepest perceptual reliance on pixel data.

8 Conclusion

We compared Early Fusion (ViLT), Mid Fusion (BLIP-2), and Token-Level Fusion (LLaVA) for low-level image quality diagnostics:

1. **Token-Level Fusion is Optimal:** Preserving 576 uncompressed patch tokens yields the highest diagnostic accuracy (67.1%) and deepest visual grounding (21.9% image gap).
2. **Query Bottlenecks Discard Detail by Default:** Mid-fusion Q-Formers discard distortion strength information due to semantic pretraining. Adapting cross-attention recovers low-level awareness ($42.4\% \rightarrow 55.3\%$), but the 32-token bottleneck caps performance on localized defects.
3. **Autoregressive Decoders are Essential:** Fixed classification heads cannot articulate low-level defects; open-ended LLMs are required to provide human-interpretable feedback for restoration networks like Restormer and NAFNet.

Future Work. Promising directions include adaptive multi-scale token allocation (dynamically allocating tokens to degraded areas while compressing uniform backgrounds) and integrating the fine-tuned VLM as an online perceptual critic for training restoration networks.

Limitations

Our study is subject to several constraints:

- **Single-Seed Runs:** ViLT and LLaVA were evaluated from single runs; BLIP-2 was trained across three seeds. With 1,495 test questions, margins under 2.5 points must be interpreted cautiously.
- **Computational Footprint:** Retaining 576 visual tokens in LLaVA increases KV-cache

Test case	Zero-shot	Fine-tuned (default)
Clean	"The image displays a picturesque agricultural landscape with hay bales under a bright, dynamic sky."	"The overall clarity is pristine with natural sharpness across the foliage, clouds, and fields. No visible compression, noise, or blurring artifacts. Quality: Excellent."
Noise (L4)	"A vintage film photograph with textured grain and dramatic lighting over open fields."	"The image exhibits severe high-frequency sensor noise and chromatic grain across the sky and foreground, noticeably reducing fine texture clarity. Quality: Poor."
Blur (L4)	"A landscape captured during misty morning dawn with ethereal atmosphere and soft lighting."	"Severe lack of edge sharpness and heavy uniform blurring obscuring foreground objects and background foliage. Focusing or motion defect. Quality: Extremely Poor."
JPEG (L4)	"A stylized scenic photograph with digital pixel grid elements across the cloud structures."	"Prominent 8×8 block boundary discontinuities, severe ringing near high-contrast edges, and visible color banding throughout the sky. Quality: Poor."
Over-smoothing (L4)	"A beautiful digital painting of hay bales in the field with smooth watercolor brush strokes."	"Unnatural loss of high-frequency texture details with plastic-like surface smoothing along edges, typical of aggressive non-linear filtering or over-denoising. Quality: Poor."
Ringing (L3)	"The quality of the image is crisp and clear with accented contour highlights along the horizon."	"Distinct Gibbs phenomenon ripples and pseudo-edges adjacent to sharp structural transitions, likely introduced by post-processing or deblurring kernels. Quality: Fair."

Table 11: Test cases: free-text quality explanations for one held-out image with a known synthetic distortion (matching Section 5.7).

Fusion Paradigm	Model	Fusion Layer	Visual Tokens	Params	Zero-Shot	Fine-Tuned	Δ Gain	Img. gap
Early Fusion	ViLT-B/32	Joint Transformer	32×32 patches	111M	47.8%	47.8%	+0.0%	–
Mid Fusion	BLIP-2 (OPT)	Q-Former Cross-Attn	32 queries	3.77B	42.4%	55.3%	+12.9%	12.8%
Token-Level Fusion	LLaVA-v1.5	2-Layer MLP	576 tokens	7.06B	**58.7%**	**67.1%**	**+8.4%**	**21.9%**

Table 12: Cross-fusion architectural comparison on Q-Bench dev ($n = 1495$). Random guessing gives 37.8%. Fine-tuned models trained on the balanced 14K Q-Instruct subset.

memory usage by $4.2\times$ during autoregressive generation compared to BLIP-2's 32 query tokens.

- **Synthetic vs. Real-World Artifacts:** Distortion sensitivity tests relied on synthetic degradations (AWGN, Gaussian blur, JPEG); real-world composite degradations in in-the-wild photography remain challenging.
- **Benchmark Scope:** Experiments were performed on the 1,495 questions of the Q-Bench dev split, as official test server ground-truth labels are held private.

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